Midterm 1 Review Worksheet

1. (True/False) Determine whether the following statements are true or false. If the statement is false, construct a counterexample that disproves the statement.

(a) The entries on the main diagonal of $A - A^T$ are all equal to 0

(b) For any two $n \times n$ square matrices A and B, $AB = BA$.

(c) For any two $n \times n$ diagonal matrices A and B, $AB = BA$.

(d) There is no 2×2 matrix A other than the identity matrix such that $A^2 = I$.

(e) If A is a skew symmetric matrix, then the elements on the main diagonal of A are all zero.

(f) Any addition of two non-invertible 2×2 matrix is non-invertible.

(g) For any two 3×3 matrices A and B, $\text{rank}(A + B) = \text{rank}(A) + \text{rank}(B)$.

2. Suppose A and C are non-singular 2×2 matrices such that

$$A^{-1} = \begin{pmatrix} 3 & 2 \\ -1 & 5 \end{pmatrix} \quad C^{-1} = \begin{pmatrix} 2 & 1 \\ -4 & 3 \end{pmatrix} \quad \text{and } \mathbf{b} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

(a) Find the solution $\mathbf{x}$ to the linear system $A^2\mathbf{x} = \mathbf{b}$.

(b) Find the solution $\mathbf{x}$ to the linear system $C^T\mathbf{x} = \mathbf{b}$.

(c) Find the solution $\mathbf{x}$ to the linear system $A^TC\mathbf{x} = \mathbf{b}$.

3. Construct nonzero 2×2 matrices A, B, and C such that

(a) $AB \neq BA$.

(b) $AB = BA$ and $A \neq B$.

(c) $AB = AC$ but $B \neq C$.

(d) $AB = 0$ but $A, B \neq 0$. (Here, 0 is the zero 2×2 matrix).

4. (True/False) Determine whether the following statements are true or false. If the statement is false, construct a counterexample that disproves the statement.

(a) If the augmented matrices of two linear systems are row equivalent, then the systems have exactly the same solutions.

(b) The reduced row echelon form of a singular square $n \times n$ matrix has a row of zeros.

(c) For any square $n \times n$ matrix A, the matrix A and its transpose A^T are row equivalent.

(d) For any 2×2 matrices A and B, $(A+B)^{-1} = A^{-1} + B^{-1}$.

5. Find all values of a for which the inverse of

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 2 & a \end{pmatrix}$$

exists. What is A^{-1}?

6. In the following linear systems, determine all values of a for which the resulting linear system has no solutions, a unique solution, or infinitely many solutions.

(a)
$$\begin{cases} x + y &= 3 \\ x + (a^2 - 8)y &= a \end{cases}$$

(b)
$$\begin{cases} x + y + z &= 2 \\ 2x + 3y + 2z &= 5 \\ 2x + 3y + (a^2 - 1)z &= a + 1 \end{cases}$$

7. (True/False) Determine whether the following statements are true or false. If the statement is false, construct a counterexample that disproves the statement.

(a) If $\det(A) = 2$, then $\det(-A) = -2$.

(b) Suppose $AB = AC$. If $\det(A) \neq 0$, then $B = C$.

(c) $\det(AB^{-1}) = \frac{\det(A)}{\det(B)}$.

(d) If $\det(AB) = 0$, then both matrices A and B are not invertible.

(e) If all the diagonal entries of an $n \times n$ matrix A are zero, then $\det(A) = 0$.

(f) If $\det(A) = 0$, then the reduced row echelon form of A has a row whose entries are all equal to 0.

(g) For any two $n \times n$ matrices A and B, $\det(A + B) = \det(A) + \det(B)$.

8. Evaluate $\det(A)$ using the cofactor expansion.

$$A = \begin{pmatrix} 5 & 6 & 0 & -1 & 3 \\ 0 & 0 & 3 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 3 & -2 & 0 & 0 & 0 \\ 0 & 2 & 3 & 1 & 0 \end{pmatrix}$$

9. Find all the values of t for which

$$\begin{vmatrix} t-1 & 0 & 1 \\ -2 & t+2 & -1 \\ 0 & 0 & t+1 \end{vmatrix} = 0.$$

10. Suppose A is an $n \times n$ matrix such that $A^T = A^{-1}$. What is $\det(A)$?

11. Let A, B, and C be 3×3 matrices such that all the entries are equal except for the 1st row. Suppose that the 1st row of C is the sum of 1st row of A and B. Use the cofactor expansion to prove that $\det(C) = \det(A) + \det(B)$.